

Chapitre 2 : Free software tools

Introduction

Software plays a central role in daily life. You use operating systems, browsers, applications, and production tools. Not all software follows the same usage rules. Different legal and philosophical models exist depending on how authors choose to share their code and their rights.

Two major movements appear:

Free software, born in the 1980s around Richard Stallman and the GNU project.

Open source, introduced in the late 1990s to make code sharing acceptable for industry.

This chapter explains fourth key ideas :

1. The **difference** between **free software** and **open source**.
2. The **types of licenses** and their legal impact.
3. The role of **Creative Commons**.
4. Exercices.

1. Difference between Free Software and Open Source:

1.1. Historical :

In the 1970s and 1980s, American universities shared their source code freely. As software became commercial, vendors started to close their code and block any modification. In response, Richard Stallman, a researcher at MIT, launched the GNU project in 1983 and created the Free Software Foundation in 1985.



The goal was to build a completely free operating system. He defined four freedoms :

1. **Use** : Use the program for any purpose
2. **Study** : Access the source code and adapt it to your needs
3. **Redistribute** : Copy and share the software
4. **Improve** : Modify the software and publish the changes

These freedoms give the user full autonomy.

Chapitre 2 : Free software tools

1.2 Rise of the Open Source movement

In 1998, figures from the free software world such as **Eric Raymond** and **Bruce Perens** created the **Open Source Initiative(OSI)**. They wanted a more practical view of code sharing without Stallman's moral and political framing.



Their goal was to convince companies that opening code improves quality, security and innovation.

This is how the term **open source** appeared, focused on transparency and technical collaboration without emphasizing ethical **freedom**.

1.3 Opposing Philosophies

The **Free Software** movement focuses on user **freedom**, ethical principles, and **the right to modify and share software**.

The **Open Source** movement emphasizes practical benefits like quality, security, and collaboration, without centering on ethics.

Key differences :

- Free Software stresses moral and social responsibility.
- Open Source stresses efficiency, innovation, and business adoption.
- Free Software uses “freedom” as a core value.
- Open Source uses “openness” as a practical strategy.

1.4 Criteria of the Open Source Initiative

The OSI defined 10 principles that a license must meet to be recognized as open source :

1. **Free Redistribution** : Software can be freely shared or sold without restrictions.
2. **Source Code Availability** : The source code must be accessible to allow modifications.
3. **Derived Works** : Users can modify the software and distribute their versions under the same license.



Chapitre 2 : Free software tools

4. **Integrity of the Author's Source Code** : Licenses may require modified versions to be clearly marked or renamed.
5. **No Discrimination Against Persons or Groups** : Anyone can use the software regardless of identity or affiliation.
6. **No Discrimination Against Fields of Endeavor** : The software can be used in any field, including commercial purposes.
7. **Distribution of License** : Rights apply to all recipients of the software without needing a separate license.
8. **License Must Not Be Specific to a Product** : The license applies to the software itself, not tied to a particular product.
9. **License Must Not Restrict Other Software** : The license cannot impose restrictions on other software distributed with it.
10. **Technology-Neutral** : The license cannot depend on a specific technology or interface.

These criteria focus on technical aspects and distribution, not on the values of freedom.

1.5 In practice

Technically, most free software is also open source, but not all open source software is considered free by the FSF.



Ask yourself

Question: Does the license let you run, study, modify and share the program without limits? If not, FSF will not classify it as free.

Answer: Software is not considered free if an open source license fails to ensure the four essential freedoms.

2. Licenses : Legal Tools for Free Software

2.1. What is a Software License ?

A **software license** is a **legal contract** through which the author allows or restricts certain uses of their work.



Chapitre 2 : Free software tools

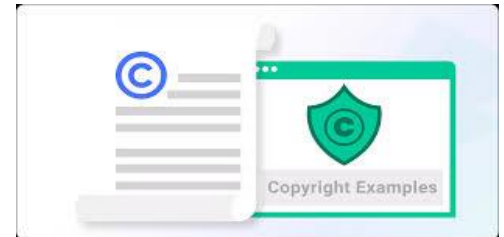
It defines :

- ✓ Copying rights
- ✓ Modification rights
- ✓ Redistribution conditions
- ✓ User obligations

Without a license, copyright applies by default : **all rights reserved**.

2.2. The Copyright Model

- ✓ **Principle:** All rights reserved
- ✓ Users cannot copy, modify, or redistribute **without permission**
- ✓ This is the model for proprietary software (Windows, Photoshop, etc.)



2.3. The Copyleft Model

Copyleft reverses copyright :

- ✓ You have the right to modify and redistribute the software
- ✓ You must preserve the same freedoms in your versions



It is a legal mechanism that ensures code freedom in all derivative versions.

a. Strong Copyleft

- ✓ All derivative works must remain under the same license
- ✓ Prevents integration into proprietary software

Examples : GNU GPL, AGPL

b. Weak Copyleft

- ✓ Only specific parts (libraries, modules) are protected
- ✓ Allows integration into other software

Examples : LGPL, Mozilla Public License (MPL)

2.4. Permissive Licenses

These licenses provide broad freedom :

- ✓ Code can be reused, modified, and integrated into proprietary software

Chapitre 2 : Free software tools

- ✓ Considered free by the FSF but not copyleft

Examples : MIT, BSD, Apache 2.0

2.5. Open Source Licenses Not Considered Free by the FSF

Some licenses approved by the OSI do not fully respect the four essential freedoms. They are open source (code is accessible) but not free because they impose usage restrictions or external obligations.

Examples :

- **JSON License** : restricts commercial use
- **Sleepycat License** : limits redistribution under certain conditions
- **Microsoft Shared Source Licenses** : allow viewing code but restrict modification or sharing

These illustrate that open source does not always equal software freedom.

3. Creative Commons (CC) Licenses

3.1. Definition

Creative Commons (CC) licenses are non-software licenses designed for cultural, educational, and artistic works (texts, images, videos, courses, etc.).

They allow authors to set the terms of use for their work:

- Attribution
- Sharing
- Non-commercial use
- No modification

3.2. The 4 Main Conditions

Symbol	Name	Meaning
BY	Attribution	Credit the author
NC	Non-Commercial	Prohibit commercial use
ND	No Derivatives	Prohibit modification
SA	ShareAlike	Derivatives must use the same license

Chapitre 2 : Free software tools

3.3. Possible Combinations

License	Meaning
CC BY	Free use with attribution
CC BY-SA	Attribution + same license (cultural copyleft)
CC BY-NC	Free non-commercial use
CC BY-ND	No modification allowed
CC0	Public domain

These licenses apply to multimedia content, not source code.

4. Office Software (LibreOffice Suite)

LibreOffice is a **free** and **open source** office suite. Includes **Writer** (word processor), **Calc** (spreadsheet), **Impress** (presentations), **Draw** (diagrams), **Base** (database), and **Math** (formula editor).

- ✓ Compatible with **Microsoft Office** formats
- ✓ **Allows** document editing, formatting, and sharing **without license costs**.
- ✓ Encourages **transparency** and **user control** over software



5. Collaboration (Storage and Sharing)

Open source tools facilitate teamwork and file management:

- ✓ **Nextcloud**: file storage, sharing, and syncing across devices
- ✓ **OwnCloud**: secure private cloud storage
- ✓ **Etherpad**: collaborative text editing in real-time
- ✓ **Mattermost** / **Rocket.Chat**: team communication and messaging



Benefits: data control, privacy, no vendor lock-in, integration with other open tools

6. Contributing to an Open Source Project

Contributing helps improve software and gain experience:

6.1 Ways to contribute:

- ✓ Reporting bugs
- ✓ Suggesting features
- ✓ Writing or reviewing code

Chapitre 2 : Free software tools

- ✓ Updating documentation
- ✓ Translating interfaces

6.2 Platforms for collaboration:

- ✓ GitHub,
- ✓ GitLab,
- ✓ SourceForge

Benefits: skill development, community recognition, networking, impact on widely used software.

7. Conclusion

The software world is structured around different legal and ethical models:

- **Free software:** focuses on **user freedom**.
- **Open source:** focuses on **code quality and transparency**

Both rely on **licenses**, but some open source licenses do **not guarantee** the four essential freedoms.

Creative Commons extends these sharing principles to **cultural content**.

Ultimately, the issue is not only technical or legal; it reflects a vision of the **digital society**, balancing **cooperation, innovation, and freedom**.

Chapitre 2 : Free software tools

8. Proposed exam

Multiple Choice

Part 1 :

1. Select the correct answer

Which term refers to a private internal network used only inside an organization?

- a) Internet, b) Intranet, c) Extranet, d) Wi-Fi

2. Which item is software?

- a) Printer, b) Router, c) Linux, d) SSD

3. Which characteristic matters most when choosing a laptop for programming?

- a) Screen color, b) RAM, c) Keyboard color, d) Logo

4. What is the main role of an operating system?

- a) Print documents, b) Manage hardware and run applications, c) Store electricity, d) Increase Wi-Fi speed

5. Which tool is used for videoconferencing?

- a) GnuCash, b) Zoom, c) ERPNext, d) Trello

6. Which advantage describes an ERP?

- a) Plays videos, b) Centralizes company operations, c) Slows down processes, d) Deletes user data

7. Which network device connects several computers in the same local area network (LAN)?

- a) Switch, b) Modem, c) Smartphone, d) Camera

8. Which example describes an IoT sensor?

- a) A mouse moving the cursor
b) A sensor measuring temperature in a smart home
c) A USB storing files
d) A webcam recording video

Chapitre 2 : Free software tools

9. Which tool allows real-time document co-editing?
- a) VLC, b) Google Docs, c) Photoshop, d) Notepad
10. What is the purpose of a workflow?
- a) Slow down the process
- b) Randomize tasks
- c) Organize steps and automate actions
- d) Delete users
11. How many essential freedoms does free software guarantee?
- a) 2 b) 3 c) 4 d) 5
12. The word “Free” in “Free Software” means:
- a) Free of charge b) Freedom c) Public d) Private
13. A copyleft license requires:
- a) Keeping the same license b) Payment c) Closing the code
14. The GPL is a:
- a) Strong copyleft b) Weak copyleft c) Permissive license

Part 2: Open-ended Questions

1. Explain the four freedoms of free software.
2. Explain the difference between Internet, Intranet, and Extranet.
3. Classify these items as hardware or software: modem, Linux, printer, Excel, Wi-Fi.
4. State two important characteristics of a laptop computer.
5. Explain the role of an operating system.
6. Give one example where you use videoconferencing in a university context.
7. Give one advantage of using an ERP in a company.
8. Identify the network device used to connect several computers inside the same laboratory.
9. Give one example of using an IoT sensor in daily life.
10. Name one collaboration tool and describe one action you can do with it.
11. Explain how a workflow can improve a process inside a university.
12. Why are permissive licenses attractive to companies?
13. What is the purpose of a Creative Commons license?